

GROZNY, Russian Federation

WMO: 372350

Lat: 43.250N

Lon: 45.717E

Elev: 258

StdP: 98.27

Time Zone: 3.00 (E03)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.6	-11.7	-16.9	0.9	-14.1	-14.2	1.1	-11.2	12.3	2.6	10.1	-1.5	1.6	290	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	33.3	21.2	31.6	20.7	30.1	20.3	22.9	30.1	22.0	28.8	21.3	27.7	3.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.9	25.9	20.0	15.1	24.9	19.2	14.4	24.3	69.5	30.5	65.8	28.7	63.2	27.9	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.3	7.1	-17.5	35.9	5.0	2.1	-21.1	37.4	-23.9	38.7	-26.7	39.8	-30.3	41.4		
			-17.8	25.5	4.9	1.8	-21.3	26.8	-24.2	27.9	-26.9	28.9	-30.5	30.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-1.4	-1.0	4.7	11.1	16.5	21.6	24.1	23.2	18.3	11.0	4.1	-0.2		
	DBStd	9.98	4.07	5.18	3.66	3.56	3.34	3.03	2.71	2.59	3.52	3.81	4.76	4.30		
	HDD10.0	1391	354	309	168	30	2	0	0	0	1	34	179	314		
	HDD18.3	3148	612	542	423	218	76	6	1	1	43	227	427	573		
	CDD10.0	1779	0	1	4	62	203	348	438	408	248	65	2	0		
	CDD18.3	497	0	0	0	1	18	104	181	151	42	1	0	0		
	CDH23.3	4281	0	0	0	13	147	872	1646	1290	304	9	0	0		
	CDH26.7	1544	0	0	0	1	20	306	668	483	66	1	0	0		
(14) Wind	WSAvg	2.6	2.2	2.2	2.8	3.3	3.1	2.6	2.7	2.5	2.4	2.4	2.4	2.0		
(15) Precipitation	PrecAvg	567	44	38	43	43	61	70	53	45	47	42	40	41		
	PrecMax	763	87	87	74	77	102	127	97	87	154	91	93	97		
	PrecMin	441	4	19	22	16	14	23	18	11	6	7	3	10		
	PrecStd	87	19	16	15	14	22	27	20	21	33	21	23	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	14.4	20.0	25.0	28.7	34.4	36.2	35.1	30.9	24.7	16.5	10.8		
		MCWB	5.8	8.6	11.7	15.2	18.8	21.4	22.6	21.4	19.9	15.9	11.4	7.3		
	2%	DB	6.1	8.8	16.0	22.5	26.7	31.9	33.8	32.7	28.5	21.6	13.4	8.1		
		MCWB	3.5	5.6	9.3	14.4	18.0	20.4	22.1	20.9	19.3	14.9	10.0	5.6		
	5%	DB	4.2	6.2	13.2	20.5	24.9	30.0	31.9	31.0	26.7	19.3	11.2	6.3		
		MCWB	2.7	3.9	8.2	13.1	17.3	20.1	20.8	20.6	18.5	13.9	8.7	4.7		
	10%	DB	2.9	4.3	10.6	18.4	23.2	28.1	30.3	29.3	24.9	17.1	9.6	4.8		
		MCWB	1.8	2.9	7.0	12.2	16.7	19.4	20.6	20.0	18.0	13.0	8.0	3.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	9.0	11.7	16.2	20.4	23.0	24.6	23.8	21.7	17.1	12.3	7.9		
		MCDB	9.0	13.0	19.6	23.2	26.7	30.9	33.2	30.3	27.6	21.5	14.2	10.5		
	2%	WB	4.1	5.8	9.8	15.0	19.2	21.6	23.5	22.5	20.5	15.5	10.4	6.2		
		MCDB	5.6	8.1	14.6	21.8	24.7	29.5	31.3	29.5	26.0	20.4	13.0	7.7		
	5%	WB	2.9	4.2	8.6	13.8	18.0	20.8	22.4	21.6	19.6	14.4	9.1	5.0		
		MCDB	3.9	5.8	12.6	19.2	23.4	28.1	29.0	28.4	24.8	18.0	10.6	5.9		
	10%	WB	2.0	3.1	7.2	12.6	17.1	20.0	21.6	21.0	18.7	13.5	8.1	3.7		
		MCDB	2.8	4.2	10.3	17.4	22.1	26.5	27.8	27.2	23.4	16.6	9.5	4.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.1	7.7	10.6	10.4	10.5	10.6	10.9	10.4	9.6	6.1	5.6		
		MCDBR	8.1	9.3	14.1	15.1	13.1	13.5	13.2	13.7	13.5	14.0	9.9	8.0		
	5% WB	MCWBR	5.8	6.0	8.0	7.5	5.7	4.9	4.3	4.3	5.6	7.3	6.2	5.6		
		MCWBR	6.8	8.5	12.9	13.4	11.7	12.1	11.5	11.9	11.4	12.2	8.7	6.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.331	0.365	0.412	0.466	0.441	0.447	0.482	0.479	0.432	0.385	0.348	0.333		
		taud	2.338	2.232	2.142	2.035	2.190	2.225	2.146	2.147	2.248	2.336	2.372	2.336		
	Ebn at Noon	Ebn at Noon	786	814	818	801	836	830	795	782	791	785	760	743		
		Edh at Noon	83	109	135	162	143	139	148	144	121	97	80	76		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.43	2.21	3.15	4.43	5.38	6.11	6.03	5.51	4.16	2.60	1.66	1.24		
		RadStd	0.21	0.40	0.38	0.52	0.56	0.39	0.47	0.44	0.42	0.34	0.23	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page